

Week 2 Report: Random Word Selection & Word Scrambling

Objective

The objective of Week 2 was to implement the random word selection mechanism and develop the word scrambling functionality. This phase focused on selecting a random word from the Word Bank and displaying it in a scrambled form to establish the core gameplay of the Word Scramble Game.

Activities Completed

1. Random Word Selection

The `java.util.Random` package was imported to generate random numbers. A random index was used to select a word from the predefined Word Bank, ensuring that a different word could be chosen each time the program runs.

2. Word Scrambling

A `scrambleWord()` method was developed to rearrange the letters of the selected word. The method converts the word into a character array, randomly swaps the characters, and returns the scrambled version of the word.

3. Program Testing

The random word selection and scrambling functions were integrated into the `Main.java` file. The program was tested several times to verify that different words were selected randomly and scrambled correctly without any errors.

Deliverables

- Updated `Main.java`
- Random Word Selection Module
- Word Scrambling Function
- Scrambled Word Display

Expected Outcome

By the end of Week 2, the program was able to randomly select a word from the Word Bank and display its scrambled version successfully. This completed the basic gameplay functionality and prepared the project for implementing player input and answer validation in the next development phase.